KEVIN LEVY

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PROFILE

Quantitative Researcher with a PhD in Astrophysics, specializing in statistical inference, machine learning, and scientific computing. Experienced building end-to-end Python and HPC pipelines, from data acquisition through statistical modeling and uncertainty quantification. Interested in research and analytical roles across academia and industry.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib), SQL, Shell Scripting
- **Data Analysis:** Data Cleaning/Validation/Visualization, Data Pipeline Development, Fourier & Time Series Analysis
- **Statistics & Machine Learning:** Frequentist & Bayesian Inference, MCMC Sampling, Supervised & Unsupervised Learning, Neural Networks
- **Cloud Platforms:** Google Cloud Platform (GCP; Vertex AI, BigQuery, Looker, Cloud Storage)
- **Infrastructure & Computing:** Linux, High-Performance Computing (HPC), Docker, Git/GitHub

PROFESSIONAL EXPERIENCE

Doctoral Researcher, Astrophysics — University of Melbourne Jun 2022 – Jun 2026

- **Scientific Impact:** Led the development of two end-to-end Python pipelines:
  1. Reconstructed and statistically analyzed the first projected matter distribution of the Universe from South Pole Telescope Summer survey data, confirming the data quality is sufficient for precise cosmological measurements;
  2. Created unbiased estimates of galaxy cluster masses from millimeter-wave data. The pipeline was later used to obtain the most constraining millimeter-wave cluster mass measurements to date (~2x precision improvement).
- **Validation**: Ran a suite of systematic tests to control for bias and ensure robust, reproducible results.
- **Collaboration & Communication**: Collaborated within an international research team on shared, version-controlled codebases and presented findings at international conferences.
- **Publications**: Levy et al. 2026, Levy et al. 2023

BI Developer — Delta IT S.A. Dec 2021 – May 2022

- **Data Engineering & Visualization**: Used SQL to build and maintain database views integrated with the WIGES ERP platform, developing dashboards and charts to track financial, buying, and selling KPIs across the client base.
- **Stakeholder Management**: Worked directly with clients on-site and via calls/Zoom to gather requirements, customizing dashboards to match each client's specific needs and preferences.

INDEPENDENT PROJECTS

Flood Detection from Satellite Imagery

- Trained a PyTorch U-Net for flood water segmentation from Sentinel-1 SAR imagery, achieving 79% mean IoU (segmentation accuracy).
- Deployed the model as a live, publicly callable API on Google Cloud Run, using a custom containerized training job on Vertex AI for scalable training.

Bayesian Model Selection for Galaxy Distance Distributions

- Built an automated Python pipeline fitting Gaussian Mixture Models to noisy galaxy distance distribution data via MCMC, using BIC for data-driven model-order selection.
- Designed an iterative reweighting scheme to jointly calibrate a bias correction across multiple data subsets.

EDUCATION

- Ph.D. in Astrophysics — University of Melbourne Jun 2022 – Jun 2026
- M.Sc. in Astrophysics — University of Bonn Oct 2019 – Oct 2021
- B.Sc. in Physics — University of Bonn Oct 2015 – Oct 2019

CERTIFICATIONS & AWARDS

- Google Cloud Computing Foundations Certificate — Google (2026)
- Data Scientist in Python Path — Dataquest.io (2021)
- N.D. Goldsworthy Scholarship (2022) — Awarded on the basis of academic excellence

LANGUAGES

English (fluent) · French (fluent) · German (fluent) · Luxembourgish (fluent)